Topics: 16.1. Vector Fields; 16.2. Line Integrals (First Part)
Please put away all electronic devices, including cell phones and calculators.

1. Sketch the vector field $\mathbf{F} = 2\mathbf{i} + x\mathbf{j}$.

2. Find the gradient vector field ∇f of f.
   (a) $f(x, y) = 2y \sin(xy)$
   (b) $f(x, y, z) = x^3 y e^{y/z}$

3. Determine whether the vector field $\mathbf{F}$ is a gradient vector field. If it is, find a function f such that $\mathbf{F} = \nabla f$. If it is not, explain why using Clairaut's Theorem.
   (a) $\mathbf{F} = 4x^3y^3\,\mathbf{i} + (3x^4y^2 - 4y^3)\,\mathbf{j}$
   (b) $\mathbf{F} = x\cos y\,\mathbf{i} + x\sin y\,\mathbf{j}$

4. Evaluate the line integral $\int_C f(x, y)\, ds$ where f and C are given.
   (a) $f(x, y) = \frac{x}{y}$, where C is given by $x = t^3$, $y = t^4$, $1 \le t \le 2$.
   (b) $f(x, y) = x + y$, where C is the upper semicircle $x^2 + y^2 = 4$ and $y \ge 0$.
   (c) $f(x, y) = xy$, where C is the arc of the curve $y = \sqrt{x}$ from $(4, 2)$ to $(9, 3)$.

5. The **average value** of a function $f(x, y)$ along a curve C is defined by

$$\bar{f} = \frac{1}{\text{length}(C)} \int_C f(x, y)\, ds.$$

   Find the average value of f along C.
   (a) $f(x, y) = \frac{x-1}{1+y^2}$, where C is the line segment from $(1, 0)$ to $(3, 1)$.
   (b) $f(x, y) = xe^y$, where C is the right half of the unit circle centered at the origin.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Vector Fields. Let D be a region in $\mathbb{R}^2$. A **vector field** on D is a function that assigns to each point (x, y) in D a vector $\mathbf{F}(x, y)$. A vector field in $\mathbb{R}^2$ has the form

$$\mathbf{F}(x, y) = P(x, y)\,\mathbf{i} + Q(x, y)\,\mathbf{j}.$$

Similarly, if E is a region in $\mathbb{R}^3$, then a vector field on E assigns to each point (x, y, z) in E a vector $\mathbf{F}(x, y, z)$. A vector field in $\mathbb{R}^3$ has the form

$$\mathbf{F}(x, y, z) = P(x, y, z)\,\mathbf{i} + Q(x, y, z)\,\mathbf{j} + R(x, y, z)\,\mathbf{k}.$$

A vector field $\mathbf{F}$ is called a **gradient field** if there exists a function f such that

$$\mathbf{F} = \nabla f.$$

In this case, $\mathbf{F}$ is also said to be **conservative**, and f is called a **potential function** for $\mathbf{F}$.

Scalar Line Integrals. Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leq t \leq b$. If f is a function whose domain contains C, then the **scalar line integral of f along C** is

$$\int_C f\,ds = \int_a^b f(\mathbf{r}(t))\,|\mathbf{r}'(t)|\,dt.$$

Suggested Textbook Problems
Section 16.1: 1-22, 25-34
Section 16.2 (First Part): 1-4, 9-12, 27, 28